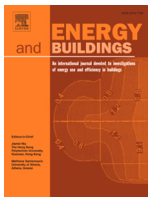
Energy & Buildings 348 (2025) 116463



Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enbuild





GUIDE: A prescriptive hybrid AI framework for energy-efficient appliances usage through behavioral modeling and LLM guidance

Alexios Papaioannou *, Asimina Dimara, Stelios Krinidis

Department of Management Science and Technology, Democritus University of Thrace, Kavala, 65404, Greece

ARTICLE INFO

Keywords:
User behavior modeling
Large language models (LLMs)
Personalized energy advice
Smart homes
Context-aware recommendations
Energy efficiency

ABSTRACT

Improving energy efficiency in residential environments reduces electricity consumption and cost, eases stress on the electricity grid, and promotes sustainability. However, realizing these benefits at scale requires accurate understanding of how energy is consumed within households. The precise interpreting of household energy consumption and providing personalized advice is still proving to be a hurdle due to overlapping appliance usage patterns and diverse user behaviors. This paper introduces Generative Usage-based Insights for Device Efficiency (GUIDE), a hybrid AI framework that integrates advanced device detection, behavioral modeling, and personalized natural language generation using fine-tuned large language models (LLMs). The system first disaggregates appliance-level consumption data utilizing statistical and temporal features, and then profiles household routines to enhance device classification accuracy. These insights are transformed into user-specific energy-saving advices through a domain-adapted LLM trained on structured behavioral prompts and contextual usage summaries. By combining time-series analysis, machine learning, and scalable LLM fine-tuning techniques (LoRA/QLoRA), GUIDE delivers fluent, context-aware and personalized recommendations that adapt to each household's evolving energy patterns. Experimental results on real-world datasets demonstrate that GUIDE achieves high accuracy in device classification and generates actionable, human-readable advice bridging the gap between signal-level analysis and personalized energy engagement.

1. Introduction

In the face of escalating climate challenges and rising energy demands, improving energy efficiency in residential settings is more important than ever. Residential energy use constitutes a significant portion of total electricity demand, with household appliances such as air conditioners, ovens, and water heaters contributing heavily to peak loads [1]. Encouraging smarter usage of these devices not only helps reduce electricity costs for consumers but also alleviates pressure on energy grids and contributes to environmental conservation. Personalized energy-saving advice tailored to user habits and device usage has proven to be more effective than generic recommendations, especially when supported by real-time data and intelligent systems [2].

The complexity of household energy consumption lies in both the diversity of devices and the idiosyncratic ways they are used. Appliances with overlapping power signatures like microwaves and electric kettles, pose challenges for accurate identification using conventional non-intrusive load monitoring (NILM) methods [3]. Furthermore, users tend to have unique habits shaped by lifestyle, seasonality, and daily routines. As a result, static or template-based feedback systems often struggle to offer advice that is truly actionable [4]. To effectively bridge the gap between raw consumption data and user-friendly guidance, energy advisory tools must evolve to interpret complex signals, contextualize usage patterns, and communicate insights in a personalized, human-centric way.

However, leveraging large language models (LLMs) for energy feedback presents its own set of challenges [5]. Most general-purpose LLMs lack domain-specific understanding of appliance-level energy consumption, user routines, or contextual nuances such as seasonality and peak demand periods. As a result, they often produce advice that is overly generic, poorly aligned with real household behaviors, or impractical for users to implement. Without proper grounding in behavioral and operational data, LLMs risk offering feedback that lacks relevance and fails to support meaningful energy reduction or efficiency improvements [6].

To address these limitations, GUIDE, a novel hybrid energy advisory framework that integrates device-level detection, behavioral modeling, and personalized natural language generation was introduced. The system begins by capturing appliance-level energy data through time-series analysis and statistical profiling. It then models household routines to

* Corresponding author.
E-mail addresses: alpapaio@mst.duth.gr (A. Papaioannou), adimara@affil.duth.gr (A. Dimara), krinidis@mst.duth.gr (S. Krinidis).

https://doi.org/10.1016/j.enbuild.2025.116463
Received 30 June 2025; Received in revised form 5 September 2025; Accepted 15 September 2025
Available online 22 September 2025
0378-7788/© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

disambiguate overlapping device signatures and extract actionable insights. A fine-tuned LLM, adapted with structured usage data, translates these insights into tailored, context-aware advice. This pipeline enables accurate device classification and the generation of human-readable, energy-saving recommendations that evolve with user behavior over time.

The proposed framework not only enhances disaggregation accuracy but also transforms energy data into personalized advice that users can readily implement. By adapting to individual behavioral profiles and seasonal usage trends, the system encourages more sustainable practices without requiring technical expertise from the user. Crucially, it achieves this by tightly integrating a fine-tuned LLM directly into the NILM pipeline, bridging signal-level analysis with natural language feedback. Unlike existing rule-based or dashboard-driven platforms, our solution provides intelligent, conversational guidance that evolves with household behavior over time offering a powerful tool for advancing user engagement and long-term energy savings.The main novelties of this paper are:

- Device classification enhanced with behavioral features, combining power consumption patterns (signal-level and statistical descriptors) with temporal usage routines to improve appliance recognition.
- Behavior-aware recommendations that move beyond generic energy feedback by delivering appliance-specific and context-sensitive insights.
- Domain-adapted LLM guidance that translates device and behavioral features into natural language, producing personalized and actionable energy-saving advices.

The remainder of this paper is organized as follows. Section 2 presents related work, covering both energy consumption behavior modeling and conversational AI systems for energy feedback. Section 3 describes the architecture of the proposed hybrid AI framework, including data collection, behavioral modeling, device detection, and the LLM-based advisory system. Section 4 details the experimental setup and evaluates the performance of both the device classification and personalized recommendation components. Finally, Section 5 concludes the paper and outlines future directions.

2. Related work

This section presents existing research across two key topic relevant to the proposed work. Section 2.1 examines prior methods in energy consumption behavior modeling and NILM within smart homes. Section 2.2 focuses on the use of conversational AI and LLMs in energy advisory systems.

2.1. Energy consumption behaviour modelling in smart homes

Several approaches have explored energy usage modeling and disaggregation using various techniques. One study employs a Markov chain probabilistic model to capture stochastic daily activity patterns and predict household energy consumption based on transitions between appliance usage states [7]. While effective for forecasting, the method abstracts user behavior into generalized states, without distinguishing between specific devices or accounting for overlapping loads. It also lacks personalized feedback, adaptation to individual routines, and scalability to dynamic usage patterns.

Another work introduces an IoT-based feedback system with visual dashboards, color-coded 3D displays, and mobile alerts designed to help users adjust consumption based on comfort preferences, achieving around 19 % energy savings over a year [8]. Despite its interactivity, the system relies heavily on user interpretation, provides limited personalization beyond preset preferences, and does not perform device disaggregation or offer language-based advisory features.

An activity-based energy demand model has also been proposed, linking occupant time-use patterns with appliance usage to analyze intraday demand heterogeneity and support demand-side management (DSM) interventions [9]. While valuable for population-level insights, this approach assumes detailed survey data and does not provide individualized, real-time feedback or direct appliance signal disaggregation.

Advances in NILM have focused on improving technical disaggregation performance. One method leverages Independent Component Analysis within a neural network framework to separate overlapping signals and reduce overfitting in multi-appliance environments [10]. Although it enhances signal-level accuracy, it omits higher-level behavioral modeling and user-facing feedback.

A deep learning approach using a residual convolutional network has also been proposed to improve recognition of low-usage appliances through long-range sequence modeling and feature extraction from smart meter data [11]. However, this technique focuses solely on appliance-level disaggregation and does not consider user behavior or feedback.

A Factorial Hidden Markov Model has been used within the Internet of Behavior framework to infer user habits from IoT-derived appliance usage data [12]. While this links usage patterns with behavioral profiling, it does not incorporate deep learning or natural language advisory components, limiting its ability to provide context-aware, personalized guidance.

Building on these appliance and behavior focused approaches, recent literature also highlights the need to situate such methods within the broader context of demand-side management and building decarbonization. Demand-side interventions are increasingly recognized as cost-effective strategies for reducing emissions and supporting low-carbon transitions. The [13] emphasizes the importance of embedding appliance-level analytics within building decarbonization pathways. Similarly, recent work in Applied Energy demonstrates how AI and machine learning can optimize demand-side flexibility in residential buildings [14].

2.2. Large language models (LLMs) in energy advisory systems

A voice-enabled, web-based assistant was developed to provide energy feedback across various timeframes, combining spoken dialogue with visual dashboards to help users understand their consumption patterns [15]. While real-world trials showed promise, the system was tested on a small scale and offered limited personalization. Its suggestions were based primarily on usage summaries, without deeper context-aware analysis or fine-grained disaggregation of similar appliances.

A rule-based text chatbot for household energy feedback demonstrated the potential of conversational interfaces using a public dataset of over 5,000 homes [16]. It provided alerts for consumption peaks via email, showing feasibility for basic interaction. However, its feedback was generic and rigid, driven by predefined rules without adapting to individual usage patterns, and lacked evaluation on user behavior or engagement outcomes.

In a simulated smart home, a voice assistant integrated with a smart speaker was designed to proactively nudge occupants toward energy-saving actions like HVAC adjustments [17]. The assistant initiated prompts without waiting for queries, showcasing an early step toward proactive advice. Yet, its scope was narrow, personalization minimal, and recommendations limited to the simulation scenario with predefined suggestions rather than dynamic, data-driven guidance.

An AI chatbot for energy systems, allowed users to query information using natural language, offering an early example of conversational AI in this domain was proposed in [18]. Its 2023 extension integrated ChatGPT, enhancing its fluency and ability to manage ambiguous queries [19]. However, both versions primarily answered general energy questions rather than providing personalized feedback based on household data. The newer model improved interaction quality but lacked

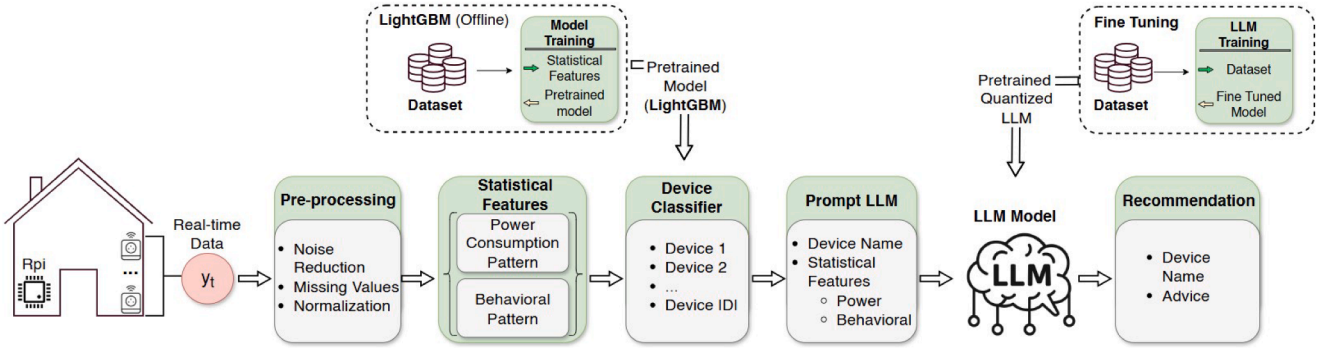


Fig. 1. GUIDE architecture for smart home device monitoring and recommendation.

user-specific context or appliance-level insights, raising concerns around relevance, accuracy, and scalability for real-world deployment.

Another approach introduced a large language model interface for Home Energy Management Systems (HEMS), which translated free-form user inputs into system configuration commands using reasoning heuristics and few-shot prompts [20]. Achieving high accuracy in interpreting user intent, the system focused on initial setup rather than continuous behavioral guidance. It did not incorporate appliance-level disaggregation or usage analytics, limiting its utility for dynamic, personalized energy management over time.

The related works reviewed in the previous sections highlight key limitations that the proposed hybrid AI framework addresses. Existing approaches often rely on abstract behavior modeling or isolated NILM techniques that fail to distinguish overlapping appliance signals or adapt to individual user routines, offering limited personalization and static feedback. Similarly, conversational systems and LLM-based assistants typically provide generic, rule-based suggestions without leveraging real household usage data or behavioral insights.

Beyond these technical constraints, most prior systems also remain disconnected from broader building-sector and demand-side objectives. While demand-side efficiency is increasingly recognized as a central lever for low-carbon transitions, existing NILM and advisory frameworks rarely translate these high-level targets into actionable, appliance-level recommendations. Furthermore, few approaches combine signal-level accuracy with behavioral interpretability, leaving a gap between what can be detected from smart meter data and what users can act upon in practice. Another key limitation is the lack of adaptive personalization: cultural context, household routines, and local tariff structures are seldom considered, restricting transferability across diverse residential settings.

In contrast, the proposed framework combines fine-grained device monitoring, user behavior modeling, and a fine-tuned LLM to generate personalized, context-aware energy advice bridging the gap between signal-level analysis and intelligent, user-specific guidance.

3. GUIDE architecture

This section presents the design and implementation of the proposed hybrid AI framework for smart energy management, the GUIDE. It describes the complete pipeline from real-time data acquisition and preprocessing to device classification and personalized energy advisory generation using a fine-tuned LLM. Each component of the system is detailed to highlight how behavioral modeling and machine learning are integrated for context-aware energy optimization.

3.1. Overview

The proposed architecture in Fig. 1 presents a complete end-to-end system for real-time monitoring, classification, and recommendation within a smart home environment using a low-power computation device (e.g. Respberry Pi) for the data acquisition. Real-time data is collected from various household devices, and then a pre-processing stage is applied, consisting of noise reduction, missing value handling, and normalization. By analyzing the processed data, statistical features are extracted including power consumption and behavioral patterns, which are passed into a pretrained LightGBM model that classifies them into specific to the device. The classification results are thus sent forward together with the associated statistical features to call in a pretrained and quantized LLM which has been fine-tuned using a customized dataset to interpret device-specific information and generate intelligent recommendations. The final output includes the identified device and actionable advice aimed at optimizing energy usage or improving operational efficiency. Therefore, this integrated approach effectively merges traditional machine learning and advanced natural language processing for smart energy management at homes.

3.2. Data acquisition and pre-processing

The architecture of the data collection system is compromized of a lightweight and modular setup of IoT devices, easily deployable and efficient in monitoring appliance-level energy consumption. Specifically, smart plugs are installed on selected household appliances to measure their real-time power usage.

The central gateway with the low-power device connects different devices and aggregates their data. The smart plugs communicate their readings using a low-range protocol (e.g. Z-wave), which is low-power, reliable wireless communication well suited for a residential environment. The low-power device is set up as an interface for a Message Queuing Telemetry Transport (MQTT) client and broker, allowing asynchronous real-time data exchange between devices. Measurement data are sampled at intervals of 1 min. These data points are stored in a structured time-series format, locally on the low-power device. This architecture, based on smart plug instrumentation, a low-power gateway, and MQTT-based messaging, supports a scalable, low-intrusion, and privacy-aware approach to home energy data acquisition.

Then, a pre-processing phase is applied to the raw data to enhance its quality. This process includes noise reduction, which removes short-duration spikes and signal anomalies caused by communication errors or hardware glitches using a moving average filter that smooths the data by averaging each point with its neighboring values. It also includes imputation of missing values, where linear interpolation and forward-filling are used to fill data gaps created by brief disconnects, and normalization, which standardizes measurement units and scales to enable uniform interpretation across different devices.

3.3. Device classification

This section presents the proposed method for device detection based on behavioral modeling. The method leverages a set of extracted statistical features, which are divided into two categories.

Table 1
Descriptions of extracted statistical behavioral features per device.

Feature name	Description
δ_i	Duration (in min) of the activation.
$Start_Hour$	Hour of the day when the pattern started.
End_Hour	Hour of the day when the pattern ended.
$Start_Hour_{Avg}$	Average start hour across all patterns on a given day.
End_Hour_{Avg}	Average end hour across all patterns on a given day.
N_d	Total number of activation patterns detected in one day.
T_d	Sum of durations for all patterns in a day.
$\bar{T}_d$	Average duration of a single activation.
Q_d	Days passed since the device was last active.
L_d	Longest consecutive number of days without usage before a current activation.
σ_d	Standard deviation of activation durations over a rolling 7-day window.
H_d	Entropy of the number of activations over a 7-day window, indicating usage variability.
$W(d)$	Sum of activations in the last 7 days (sliding window).
$W_T(d)$	Average daily active duration over the past 7 days.
$U(d)$	Number of days with at least one activation over the last 7 days.

Table 2
Extracted statistical features from power signal.

Feature	Description
$P_1, P_2, \ldots, P_q$	Top 5 dominant periods (in min) based on the inverse of the highest peaks from the power spectral density (Periodogram).
$CycleLen$	Total number of samples in the activation cycle.
$ACF_1, ACF_2, \ldots, ACF_q$	Autocorrelation coefficients at lags 1 through 5; measures linear dependence over time.
$PACF_1, PACF_2, \ldots, PACF_q$	Partial autocorrelation coefficients at lags 1 through 5; isolates correlation at each lag.
$Mean$	Arithmetic mean of the power values in the activation.
Std	Standard deviation of the power signal.
$Skew$	Asymmetry of the power distribution.
$Kurt$	Peakedness or flatness of the power distribution compared to a normal distribution.
$HMean$	Harmonic mean of non-zero power values, emphasizing smaller values.
$CrestF$	Ratio of peak power to mean power.
$NumPeaks$	Number of local peaks identified in the power signal.
$PeakProm_{Avg}$	Average prominence (vertical distance to surrounding minima) of all detected peaks.
$PeakProm_{Max}$	Maximum prominence among all peaks.
$PeakW_{Avg}$	Average width of peaks at half height.
$PeakW_{Max}$	Maximum width of peaks at half height.
$PeakDist_{Avg}$	Average horizontal distance between consecutive peaks.
$PeakDist_{Max}$	Maximum horizontal distance between consecutive peaks.
$PeakAmp_{Avg}$	Average difference between adjacent peak values.
$PeakAmp_{Max}$	Maximum difference between adjacent peak values.
$PeakHt_{Avg}$	Average height of peaks (value at local maximum).
$PeakHt_{Max}$	Maximum height among detected peaks.

3.3.1. Feature extraction

The first category captures behavioral dynamics over time and is summarized in Table 1. Those data are important as they represent core behavioral dimensions of residential appliance usage. In NILM and behavioral modeling literature, temporal and duration-based descriptors are widely recognized as effective indicators of household routines and user habits. The start and end time of a device are important for distinguishing appliances with overlapping load signatures but distinct daily usage periods [21]. Duration encapsulates regularity and irregularity of operation, allowing for separation between short from long cycles. Frequency and variability-based feature representations can help to reveal weekly patterns and disorderly behaviours in usage that are consistent with established energy behavior modeling techniques [22].

The second group of features focuses on the power consumption pattern characteristics of individual activations, which are detailed in Table 2. The features listed above are calculated directly from the high-resolution power measurements taken by the smart plugs in the system. Each activation event is segmented from the continuous time-series, and its raw power signal is then used to extract statistical descriptors. Frequency-domain features from the sampled signal (e.g. periodogram peaks), as well as autocorrelation and partial autocorrelation values, can be taken directly from the temporal activation sequence. Distributional statistics (mean, standard deviation, skewness, kurtosis) and peak morphology descriptors (prominence, width, amplitude, distance) are obtained by applying standard signal-processing algorithms to the same activation traces. These methods do not require any additional hardware beyond smart plugs and sampling at 1-min resolution, ensuring feasibility in real-world household deployments.

To comprehensively characterize device usage behavior, several temporal and statistical behavioral features were extracted from daily appliance activity. These features aim to capture frequency and variability in usage patterns, which are critical for understanding routine and non-routine energy-consumption behavior.

Initially, *the number of activations per day* is extracted, denoted N_d, representing the count of activation patterns detected for a specific appliance on a day d. The corresponding *total active duration* for that day, T_d, is defined as the total duration in minutes of all activation patterns and computed as follows:

$$T_d = \sum_{i=1}^{N_d} \delta_i, \tag{1}$$

each activation event is indexed by i and the duration by δ_i

The *average activation duration* $\bar{T}_d$, which signifies a usual period of use for an activation event, is computed as:

$$\bar{T}_d = \begin{cases} \frac{T_d}{N_d} & \text{if } N_d > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

To calculate the periods of inactivity, two metrics were introduced. The *days since last use*, defined as Q_d which describes the duration (measured in days) since the appliance was last activated at the current day, day d. Similarity the *longest inactive streak prior to a given activation,* defined as L_d, indicates the largest number of consecutive days during

which the device was not in use before it was again activated. Both features are designed to highlight irregular usage and extended periods of inactivity.

A 7-day rolling window is calculated in order to determine the standard deviation of the daily average activation durations over time, to evaluate the consistency of appliance use across time interval points. This rolling window of 7 days was selected in order to capture variance in appliance usage. This horizon corresponds closely to the natural rhythm of residential life. Appliance usage typically differs for residents throughout the week compared to the weekend. A rolling window avoids defining specific calendar weeks in feature extraction, and thus avoids artificially resetting at week boundaries. The 7-day rolling window also offers a trade-off between sensitivity and stability, as it is long enough to capture routine weekly cycles, but short enough to remain sensitive to short-term changes in user behavior. Longer horizons like monthly, would dampen these fluctuations and tend to mask important short-term variations in appliance use. This feature, referred to as *daily usage variability* and denoted σ_d:

$$\sigma_d = \sqrt{\frac{1}{7}\sum_{i=0}^{6}\left(\bar{T}_{d-i} - \mu_d\right)^2} \tag{3}$$

where μ_d is the mean of the 7-day average activation durations.

Additionally, entropy is used to quantify the unpredictability in daily activation patterns. The entropy of appliance usage over a 7-day period ending at day d is expressed as:

$$H_d = -\sum_{z=1}^{k} p_z \log(p_z), \tag{4}$$

where p_z represents the relative frequency that a day within the 7-day window had exactly z activations and k refers to the number of activations observed on any single day within the 7-day window. High entropy values correspond to highly irregular and diverse usage patterns, while low values reflect consistent behavior.

Weekly aggregation features were also extracted to capture medium-term trends. The *total number of activations during this period* is denoted by $W(d)$:

$$W(d) = \sum_{j=d-6}^{d} N_j. \tag{5}$$

Similarly, the *weekly average activate duration*, $W_{\bar{T}}(d)$, exhibits the mean daily active use across the window:

$$W_{\bar{T}}(d) = \frac{1}{7}\sum_{j=d-6}^{d} T_j \tag{6}$$

Finally, the *count of unique usage days within the window* is denoted as $U(d)$ and refers to the number of activation days within the seven days:

$$U(d) = \sum_{j=d-6}^{d} \frac{\text{sign}(N_j)+1}{2} \tag{7}$$

where

$$\frac{\text{sign}(N_j)+1}{2} = \begin{cases} 1, & \text{if } N_j > 0 \\ 0, & \text{otherwise} \end{cases}$$

Together, these behavioral features offer a strong temporal profile regarding the usage of appliances, collecting attributes of frequency and variability in daily and weekly routines.

To extract useful information from the power consumption pattern of each device, a set of statistical features was extracted. These features were expected to represent conditions of power consumption distribution, and peak morphology of the pattern.

Initially, the frequency characteristics of each activation were analyzed using the periodogram, which estimates the power spectral density of the signal. The q most dominant periodic components were identified by locating the highest spectral peaks. These are converted into time-domain periods, denoted as $\mathrm{P}_1, \ldots, \mathrm{P}_q$, and are computed as:

$$\mathrm{P}_r = \frac{1}{f_r}, \tag{8}$$

where f_r is the frequency corresponding to the r-th peak in the spectrum. These values help capture repetitive or cyclic activity in the appliance operation.

Following this, to model temporal dependencies in the power consumption signal, the autocorrelation function (ACF) and partial autocorrelation function (PACF) were computed up to lag q. Let $s = \{s_1, s_2, \ldots, s_n\}$ denote the sequence of power measurements within an activation pattern, where n is the total number of samples in the sequence, and $\bar{s}$ is the mean of the signal. The ACF at lag k, denoted ρ_k, quantifies the linear dependence between the signal and its lagged version and is defined as:

$$\rho_k = \frac{\sum_{t=1}^{n-k}(s_t - \bar{s})(s_{t+k} - \bar{s})}{\sum_{t=1}^{n}(s_t - \bar{s})^2} \tag{9}$$

Basic statistical features of the signal include the μ, standard deviation (σ), skewness (γ), and kurtosis (κ), defined respectively as:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} s_i, \quad \sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(s_i - \mu)^2} \tag{10}$$

$$\gamma = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{s_i - \mu}{\sigma}\right)^3, \quad \kappa = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{s_i - \mu}{\sigma}\right)^4 \tag{11}$$

Moreover, the harmonic mean was included to emphasize smaller power values while discounting large peaks:

$$H = \frac{n}{\sum_{i=1}^{n}\frac{1}{s_i}}, \quad s_i \neq 0 \tag{12}$$

Additionally, the crest factor provides a measure of peak sharpness and is computed as the ratio of the maximum power to the mean power:

$$C = \frac{\max\{s\}}{\mu + \varepsilon}, \tag{13}$$

where ε is a small constant added to avoid division by zero.

Finally, local peaks in the signal were extracted using a peak detection algorithm, producing metrics such as the number of peaks, mean and maximum peak prominence, width, distance, amplitude, and height.

The combined set of daily and weekly behavioral features, along with the statistical properties of the power signals, provides a robust profile for distinguishing between different appliances and supporting precise device identification.

3.3.2. Classification method

To utilize the extracted features for device recognition, a LightGBM (Light Gradient Boosting Machine) classifier is used. LightGBM constructs an ensemble of decision trees to perform multi-class classification of the devices.

Given a feature vector $\mathbf{x}_i$ for the i-th sample, the LightGBM classifier predicts the corresponding device class y_i by minimizing the following objective function:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, F(\mathbf{x}_i)) + \sum_{k=1}^{K}\Omega(f_k), \tag{14}$$

where K is the total number of decision trees in the model, $l(y_i, F(\mathbf{x}_i))$ is the loss function (typically the multi-class cross-entropy loss), $F(\mathbf{x}_i)$ is the model prediction obtained by the sum of all weak learners f_k, and $\Omega(f_k)$ is the regularization term to prevent overfitting.

The LightGBM model updates iteratively by adding new decision trees that correct the residual errors of the ensemble. The optimization

at each iteration involves minimizing the Taylor-expanded approximation of the objective function:

$$\text{gain} = \frac{1}{2}\left[\frac{\left(\sum_{i\in I_L} g_i\right)^2}{\sum_{i\in I_L} h_i + \lambda} + \frac{\left(\sum_{i\in I_R} g_i\right)^2}{\sum_{i\in I_R} h_i + \lambda} - \frac{\left(\sum_{i\in I} g_i\right)^2}{\sum_{i\in I} h_i + \lambda}\right], \tag{15}$$

where g_i and h_i are the first and second derivatives (gradients and Hessians) of the loss with respect to the model output, I_L and I_R represent the sets of samples assigned to the left and right child nodes respectively, $I = I_L \cup I_R$ is the set of samples at the current node before splitting, and λ is a regularization parameter.

3.4. User behavior patterns

Home appliances exhibit varying patterns of usage, influenced significantly by user behavior and lifestyle. Thus, appliances can be categorized and analysed according to the usage behavior by using two significant types: Constantly-On appliances and On-Demand appliances [23].

Constantly-on devices continuously operating or maintained in standby mode without any users intervention. Their energy consumption remains relatively constant throughout the day, typically with minor fluctuations. Examples of such appliances include: refrigerators and freezers, routers and modems as well as security systems.

On the other hand, On-Demand appliances are enabled based on user needs and usage schedules. Their consumption profiles are therefore more variable depending directly on the user's lifestyle, activities, and preferences. This category is further subdivided into two sub-categories, on user interaction complexity: Program-Based and Non-Programmed Appliances.

Program-based devices operate based on predefined or user-programmed cycles, typically running specific routines or tasks. Examples include: Washing machines, Dishwashers and Clothes dryers. The usage behavior of program-based appliances displays predictable and repetitive patterns. Users usually switch on these appliances at certain times (e.g., at off-peak electricity hours, on weekends, or in line with a daily routine), influenced by habits, convenience, or economic factors.

Non-Program Based appliances operate without programmed cycles or automatic timing. They are typically activated spontaneously according to immediate user requirement. Examples include: Televisions, microwaves, ovens, electric kettles and hairdryers. The energy consumption pattern for these devices is irregular and highly variable as they are characterized by an unpredictable frequency, duration, and intensity of use, which would primarily rely on user habits, routines, personal schedules, as well as external environmental factors such as seasonal variations or daylight availability.

Each home appliance exhibits distinct operational characteristics such as power consumption levels, duration, and frequency of use creating unique identifiable usage patterns [24]. These signatures can be effectively used for device recognition and disaggregation. However, certain appliances demonstrate notably similar usage patterns, complicating their accurate identification based solely on electrical signatures [25]. However, these ambiguities can typically be resolved by incorporating contextual information derived from user behavior, such as the typical timing of appliance usage or common sequences of appliance activation. Analyzing such behavioral patterns enables accurate differentiation between appliances with overlapping or closely related electrical signatures.

3.5. LLM-based advisory system

The final component of the proposed framework leverages the capabilities of LLMs to generate intelligent, personalized recommendations aimed at optimizing household energy consumption. Traditional advisory systems often lack contextual understanding and fail to adapt to user-specific behavior, resulting in generic suggestions with limited impact. In contrast, the proposed LLM-based advisory system utilizes structured behavioral data derived from appliance usage to produce context-aware, advice. By fine-tuning the LLM over hand-crafted prompt-response pairs, the system is capable of understanding complex usage patterns and generating real-time actionable insights. This section outlines the processes involved in dataset preparation, model selection, instruction tuning, parameter-efficient fine-tuning techniques, and prompt engineering are all aimed at developing a scalable and responsive energy guidance engine.

3.5.1. Dataset creation

To enable effective instruction-tuning of the LLM, a specific dataset was constructed from smart home appliance. Unlike generic language datasets, this one focused on detailed behavioral insights derived from real-world device usage.

Smart plugs were used to monitor a range of household appliances, capturing time-stamped energy data such as activation periods, power levels, and runtime durations. From this raw data, structured profiles were built for each device on a daily basis. These profiles included statistical and temporal features described in Section 3.3.

In addition, dataset with a small domain corpus drawn from publicly available energy-efficiency reports, appliance usage manuals, household energy-saving guidelines ([26,27], [28]) academic and industry publications on behavioral energy saving, demand-side management, and appliance usage routines was added [29]. These documents were pre-processed into instruction–response pairs that provide generalizable templates for recommendations, ensuring that the model learns both from structured device statistics and broader domain knowledge.

To generate corresponding advisory outputs, a rule-based labeling mechanism was applied. Behavioral thresholds specific to each appliance type were used to classify usage patterns into qualitative categories including normal, excessive, or sporadic.

Instead of manually writing each recommendation, a set of predefined templates was used. These templates produced natural language suggestions based to the observed behavior, covering areas such as efficiency improvements, usage regularity, and wear prevention. These templates were organized into three intent categories:

- Efficiency advice, e.g., "Your [device] often runs during peak tariff hours; shifting use to off-peak times may lower costs."
- Regularity feedback, e.g., "Your [device] shows irregular usage patterns; establishing a routine schedule could improve efficiency."
- Maintenance guidance, e.g., "Frequent use of the [device] may increase mechanical wear; consider reducing activation frequency when possible."

Each template had multiple using variations to increase diversity, including paraphrases and optional clauses tailored to the device type. Templates were parameterized so that placeholders (e.g., [device], [usage pattern]) could be automatically filled from the behavioral labels, producing outputs that were both context-aware and varied in style.

Each dataset entry was then formatted as a prompt response pair. Prompts followed a structured schema containing the device type, summary statistics (e.g., activations per day, average duration, total runtime), and behavioral context labels. The paired response was a natural-language advisory message drawn from the template bank, aligned with the detected behavior. This ensured that the LLM would learn consistent mappings from structured appliance features to actionable recommendations. Fig. 2 presents the whole process.

3.5.2. Instruction-tuning methodology

To equip the LLM with the capability to generate energy-saving recommendations based on behavioral of the users, an instruction-tuning approach was developed. This process involved supervised fine-tuning on a custom-crafted dataset composed of prompt-response pairs, where

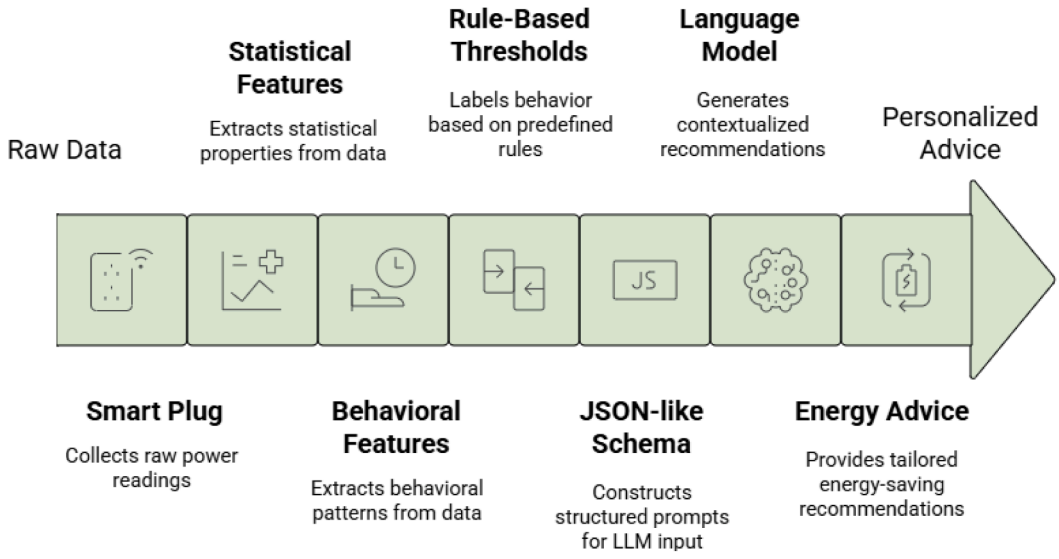


Fig. 2. GUIDE process flow from time-series data to LLM-generated recommendations.

each prompt encodes a structured summary of an appliance's usage profile, and the response provides a human-readable, context-aware advisory message.

Each prompt included key behavioral features:

- Device name
- Activations per day
- Average activation duration
- Total runtime
- Days since last use
- Daily usage variability (standard deviation)
- Usage entropy (pattern irregularity)

An example prompt format:

```
Device: Electric Heater
Activations per day: 4
Average activation duration: 75\,min
Total active duration: 300\,min
Days since last use: 0
Usage standard deviation: 2.8
Usage entropy: 1.2
```

Responses were formulated based on predefined rules connected to feature thresholds (e.g., low, typical, high usage). They provided practical advice addressing three core dimensions:

- Energy Efficiency: Suggestions for batching usage, reducing idle consumption, or optimizing schedules.
- Prescriptive Maintenance: Alerts on overuse that could signal wear or risk of overheating and breaking down.
- Operational Consistency: Encouragement for more predictable usage to extend appliance lifespan and improve efficiency.

Example response:

> *Usage of the electric heater appears high today. Consider lowering activation frequency or optimizing usage during off-peak hours. Establishing a regular heating schedule may help reduce energy costs and device strain.*

The instruction-tuning was formulated as a conditional language modeling task. The LLM was trained to generate accurate and diverse advisory responses conditioned on input prompts. Loss was computed using standard cross-entropy between generated and target tokens.

To avoid repetitive output and improve robustness, multiple linguistic variations of each advice template were introduced. Synonyms, sentence reordering, and alternative phrasings were used to expose the model to a wider range of expression styles. This tuning approach allowed the LLM to learn both the semantic relevance of energy behavior features and the pragmatic tone needed for effective end-user communication.

3.5.3. Low-rank adaptation (LoRA) and quantized low-rank adaptation (QLoRA)

In traditional full fine-tuning, updating all the parameters of a LLM such as those found in Transformer architectures is computationally intensive and memory-demanding. To mitigate this, parameter-efficient fine-tuning techniques have been introduced, among which Low-Rank Adaptation (LoRA) [30] and Quantized Low-Rank Adaptation (QLoRA) [31] have emerged as a widely adopted method.

By inserting trainable low-rank matrices into particular network layers, such as the self-attention projections, LoRA avoids having to update the entire model. Let $W_0 \in \mathbb{R}^{d^{(l)} \times k^{(l)}}$ be a pre-trained weight matrix in the transformer block, where $d^{(l)}$ is the output dimension of the model layer, and $k^{(l)}$ represents the input dimension of the feature space. which transforms an input vector $x^{(l)} \in \mathbb{R}^{k^{(l)}}$ into an output $h^{(l)} \in \mathbb{R}^{d^{(l)}}$ through the standard operation $h = W_0 x^{(l)}$. LoRA modifies this operation as follows:

$$h = W_0 x^{(l)} + BAx^{(l)} \tag{16}$$

where: $A \in \mathbb{R}^{r \times k^{(l)}}$ and $B \in \mathbb{R}^{d^{(l)} \times r}$ are learnable low-rank matrices, $r \ll \min(d^{(l)}, k^{(l)})$, ensuring that the total number of trainable parameters $d^{(l)}r + rk^{(l)}$ is significantly smaller than $d^{(l)}k^{(l)}$, W_0 remains frozen during training, and only A and B are updated.

This low-rank adaptation is generally applicable to any module within the LLM where parameter updates are desired, such as query, key, and value projections in the attention mechanism.

The training objective for LoRA-based fine-tuning, similar to that used in full fine-tuning, seeks to maximize the log-likelihood of the output tokens given the inputs. Let Θ_{pre} be the set of parameters of the frozen pre-trained model and $\Delta\Theta = \{A_l, B_l\}_{l=1}^{L}$ the set of trainable LoRA parameters across L layers. The objective function can be formulated as:

$$\max_{\Delta\Theta} \sum_{u} \frac{1}{|y_u|} \sum_{t=1}^{|y_u|} \log P_{\Theta_{\text{pre}}+\Delta\Theta}(y_t^u \mid y_{<t}^u, x_u^{(l)}) \tag{17}$$

where y_u represents the target output sequence for a given input $x_u^{(l)}$, $|y_u|$ is the length of the sequence y_u, $P_{\Theta_{\text{pre}}+\Delta\Theta}(y_t^\mu \mid y_{<t}^\mu, x_u^{(l)})$ represents the probability of generating token y_t^μ given previous tokens $y_{<t}^\mu$ and input $x_u^{(l)}$, Θ_{pre} are the pre-trained model parameters, $\Delta\Theta$ are the updated parameters introduced via LoRA, modifying specific weight matrices without altering the entire model, $y_{<t}^\mu$ refers to all previous tokens up to time step t and $x_u^{(l)}$ is the conditioning input for the model.

Despite LoRA successfully lowers the number of trainable parameters during fine-tuning, the base model weights are still computed and stored in full-precision. This poses challenges in terms of memory and scalability when applied to extremely large models. To further optimize memory usage and enable fine-tuning on consumer-grade hardware, QLoRA has been proposed as a memory-efficient extension of LoRA.

QLoRA builds on the foundation laid by LoRA by integrating 4-bit quantization of the base model weights, as outlined in (18). Specifically, instead of directly using the full-precision weight matrix $W_0 \in \mathbb{R}^{d^{(l)}\times k^{(l)}}$, QLoRA uses a quantized version $W_0^q \in \text{INT4}^{d^{(l)}\times k^{(l)}}$, which is dequantized on-the-fly during the forward pass. The standard linear transformation $h^{(l)} = W_0 x^{(l)}$ in (16) is replaced by:

$$h^{(l)} = \text{dequant}(W_0^q)x^{(l)} + BAx^{(l)}. \tag{18}$$

Only A and B are trainable while W_0^q remains fixed. The number of trainable parameters introduced by the adapter is $d^{(l)}r + rk^{(l)}$, which is significantly smaller than the full parameter size dk, ensuring parameter-efficiency.

QLoRA enhances low-rank adaptation and quantization, integrating advanced system-level optimizations. These include double quantization to further compress quantization constants, optimizing memory usage. Additionally, paged optimizers enhance memory access patterns during training in hardware constrained environments. These innovations enable QLoRA to efficiently adapt large pre-trained models to downstream tasks with minimal computational overhead and memory footprint, while maintaining the performance advantages of full-parameter fine-tuning.

Both LoRA and QLoRA enable scalable and efficient fine-tuning of large models, making them highly suitable for research and deployment on resource-constrained hardware.

4. Experimental results

This section presents the experimental evaluation of the proposed hybrid AI framework for energy-efficient appliance usage, focusing on both the accuracy of device detection and the effectiveness of the LLM-based advisory system in generating personalized energy-saving recommendations. The evaluation aims to validate the system under realistic operational conditions using a public dataset.

4.1. Experimental setup

The experiments were conducted using the REFIT Electrical Load Measurements dataset, a publicly available resource that provides detailed appliance-level energy consumption data collected from multiple households in the United Kingdom [32]. REFIT offers high-frequency power readings at 1-min intervals, making it an ideal testbed for residential load disaggregation and behavioral modeling studies. It consists of smart plug data collected from 20 UK households, mainly detached or semi-detached homes, occupied by families of varying sizes, over a period of 2013–2015 (two years), with each household monitoring between 8 and 20 appliances. From the REFIT dataset, a subset of commonly used household devices was selected for experimentation, including a fridge, washing machine, dishwasher, clothes dryer, television, kettle, microwave, toaster, and electric heater. This appliance selection covers a diverse range of usage behaviors, including both constantly-on and on-demand devices, thereby providing a robust basis for evaluating the system's generalizability.

All model training and fine-tuning activities were performed on Google Colab. The training leveraged Colab's A100 GPU, with a runtime environment consisting of Python 3.10, PyTorch 2.0, LightGBM 3.3.2, and the HuggingFace transformers library. The quantized fine-tuning of large language models was facilitated using the bitsandbytes package, supporting efficient low-precision matrix operations necessary for QLoRA.

Data preprocessing included the standard operations described in Section 3.2, such as noise reduction through smoothing filters, imputation of missing values via linear interpolation, and normalization of power signals. The LightGBM classifier was trained with a learning rate of 0.05, 64 maximum leaves, and 500 boosting rounds, with early stopping triggered after 50 consecutive non-improving rounds. For the advisory system, the LLM selected for fine-tuning were a 1.5B parameter version of DeepSeek, ChatGPT2 and Qwen with 1.8B parameters. Fine-tuning implemented LoRA and QLoRA configurations with rank (8,16,32), alpha (16,32,64), and a dropout rate of 0.1, using a batch size of 32 and a sequence length of 512 tokens. This environment allowed for rapid experimentation without the need for dedicated local servers, ensuring accessibility and reproducibility. All information is presented in Table 3.

4.2. Evaluation metrics

To ensure a comprehensive evaluation of the proposed system across device classification and natural language generation tasks, a set of established performance metrics was utilized, each selected based on its mathematical suitability for the specific objective.

Accuracy (Acc) evaluates the proportion of exactly matched outputs:

$$\text{Acc} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} \tag{19}$$

Although strict for natural language tasks, it provides an upper bound on advisory generation precision.

The F1-Score (F_1) balances precision and recall, especially relevant for both device classification and natural language output quality:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{20}$$

The perplexity (PPL) quantifies the language model's uncertainty and is computed as:

$$\text{PPL} = \exp\left(-\frac{1}{N}\sum_{i=1}^{N} \log p(y_i|y_{<i}, x)\right) \tag{21}$$

where $p(y_i|y_{<i}, x)$ represents the probability of generating the token y_i given the previous tokens $y_{<i}$ and the input prompt x, N is the number of tokens, y_i the predicted token, and x the input prompt. Lower perplexity indicates better model fluency.

ROUGE-1 and ROUGE-L were employed to assess lexical and structural overlap between generated advice and ground-truth responses. ROUGE-1 measures unigram overlap, while ROUGE-L captures the longest common subsequence, focusing on the preservation of coherent structure. Given R (reference) and C (candidate), ROUGE recall is formulated as:

$$\text{ROUGE-1 Recall} = \frac{|\text{Unigrams}(R) \cap \text{Unigrams}(C)|}{|\text{Unigrams}(R)|} \tag{22}$$

where Unigrams is the set of all words (tokens) in the reference text, which is the ground-truth output. Similarly the ROUGE-L is calculated based on longest subsequence matching.

Finally, BERTScore computes semantic similarity between generated and reference outputs using contextual embeddings $e(\cdot)$ from a pre-trained BERT model:

$$BERTScore = \frac{e(C) * e(R)}{||e(C)||||e(R)||} \tag{23}$$

where $e(C)$ and $e(R)$ represent the embedding vectors of the candidate text (C) and the reference text (R), respectively.

Table 3
Summary of experimental setup and model parameters.

Aspect	Details
Input Features	Extracted Behavioral Features (Table 1) and signal-level features (Table 2)
Output Variable	Appliance class (device label)
Dataset	REFIT Electrical Load Measurements
Appliances Selected	Fridge, Washing Machine, Dishwasher, Dryer, TV, Kettle, Microwave, Toaster, Electric Heater
Sampling Rate	1 min intervals
Data Preprocessing	Noise reduction, missing value imputation, normalization
Training Environment	Google Colab (A100 GPU)
Software Stack	Python 3.10, PyTorch 2.0, LightGBM 3.3.2, HuggingFace Transformers, bitsandbytes
LightGBM Configuration	Learning Rate: 0.05; Max Leaves: 64; Boosting Rounds: 500; Early Stopping: 50
LLM Base Model	DeepSeek 1.5B, ChatGPT2, Qwen 1.8B
Fine-tuning Method	LoRA / QLoRA
LoRA / QLoRA Settings	Rank: 16; Alpha: 32; Dropout: 0.1
Batch Size	32
Epoch	5
Sequence Length	512 tokens

The chosen metrics reflect the dual objectives of the system, accurate device detection through statistical metrics (Accuracy, F1-Score) and contextually faithful, linguistically fluent recommendations through advanced generative metrics (Perplexity, ROUGE, BERTScore).

4.3. Device detection evaluation

In this paper, the classification task involved nine appliance classes: fridge, washing machine, dishwasher, clothes dryer, television, kettle, microwave, toaster, and electric heater. These appliances were selected from the REFIT dataset as they occur frequently in households, as well as representing both constantly-on (e.g., fridge) and on-demand appliances (e.g., kettle, microwave). The input variables consisted of the behavioral characteristics detailed in Table 1 (e.g., counts of activations, start/end times, variation of use, weekly frequency) and the signal-level characteristics detailed in Table 2 (e.g., periodogram peaks, autocorrelation, statistical metrics of the distribution, peak moral forms). The output variable was the appliance class label (Table 3).

The LightGBM model was trained on the combined statistical and behavioral feature sets extracted from the REFIT dataset, covering temporal dynamics, activation patterns, and power signature characteristics as detailed in Section 3.3. Device classification performance was assessed through evaluation metrics, including precision, recall, and F1-Score.

To comprehensively evaluate the effectiveness of the proposed approach, a comparative study was conducted against several classical State-of-the-Art (SoA) machine learning models commonly applied in non-intrusive load monitoring and general classification tasks. The baseline models included Support Vector Machine (SVM), XGBoost, Random Forest (RF), Decision Trees (DT), and AdaBoost. For each model, two training settings were considered: one using only statistical features extracted from the raw power consumption signals, and another enhanced with behavioral modeling features.

The results are summarized in Table 4. As shown, across all models, the inclusion of behavioral features consistently improved classification performance across all evaluation metrics. In particular, the proposed LightGBM model with behavioral modeling achieved the highest scores, recording a precision of 0.9210, recall of 0.9060, and an F1-Score of 0.9130. This represents a significant improvement over the LightGBM model trained without behavioral features, which achieved an F1-Score of 0.7950.

Among the classical models trained without behavioral features, XGBoost and Random Forest showed relatively strong performance, achieving F1-Scores of 0.7700 and 0.7800, respectively. However, even these models benefited from behavioral information, improving their F1-Scores to 0.8760 and 0.8690 respectively after integrating behavioral features. In contrast, Decision Trees consistently demonstrated the lowest performance, with an F1-Score of 0.7600 without behavioral features, highlighting their tendency to overfit in complex feature spaces when used without ensemble techniques.

Table 4
Device detection performance: with and without behavioral features (averaged over all appliances).

Model	Precision		Recall		F1-Score	
	Without	With	Without	With	Without	With
DT	0.7700	0.8550	0.7500	0.8380	0.7600	0.8465
RF	0.7900	0.8780	0.7700	0.8600	0.7800	0.8690
AdaBoost	0.7700	0.8690	0.7500	0.8530	0.7600	0.8610
SVM	0.7650	0.8600	0.7450	0.8420	0.7550	0.8510
XGBoost	0.7800	0.8840	0.7600	0.8680	0.7700	0.8760
LightGBM	0.8050	**0.9210**	0.7850	**0.9060**	0.7950	**0.9130**

AdaBoost and SVM exhibited moderate classification capabilities, with F1-Scores of 0.7600 and 0.7550 respectively in the absence of behavioral features. Upon integrating behavioral modeling, these scores increased to 0.8610 and 0.8510, respectively, further reinforcing the value of contextual behavioral data in improving device disambiguation tasks. Detailed results for each appliances and algorithms are provided in Table 6.

Overall, these findings underline that while strong baseline classifiers can achieve reasonable accuracy based purely on statistical signals, the integration of behavioral patterns provides an important increase in all evaluation metrics. As shown in Table 6, the LightGBM model with behavioral features achieved the highest overall F1-score of 0.913, outperforming Decision Trees, Random Forest, AdaBoost, SVM, and XGBoost baselines, which are commonly applied in NILM-related classification tasks.

4.4. Advisory system evaluation

The evaluation of the LLM-based advisory system focused on quantitative and efficiency measures to consider the performance and scalability characteristics of the various fine-tuning configurations. Two adaptation mechanisms were compared: LoRA and QLoRA, each evaluated under two hyperparameter settings ($r = 8$, $\alpha = 16$, $r = 16$, $\alpha = 32$ and $r = 32$, $\alpha = 64$).

Table 5 presents the detailed results. Overall, DeepSeek demonstrated the best performance among all models. Under LoRA fine-tuning, it achieved accuracy values of 0.892, 0.881, and 0.903 across configurations, with corresponding F1 scores of 0.899, 0.883, and 0.9007. The ROUGE-1 and ROUGE-L values were between 0.3037 and 0.3361 while BERT score was high, ranging between 0.8447 and 0.8483, ensuring semantic smoothness. Perplexity values remained low and stable, from 1.13 to 1.14, indicating grammatical coherence in output. Under

Table 5
LLM advisory system performance across different LoRA and QLoRA configurations.

Models	Tuning method	r	α	Per.	Acc.	ROUGE-1	ROUGE-L	F1 Score	BERT score	Inference time (s)	Size (GB)
DeepSeek	LoRA	8	16	1.14	0.892	0.3274	0.256	0.899	0.8447	2.01	3.31
	LoRA	16	32	1.13	0.881	0.3343	0.2546	0.883	0.8476	2.01	3.31
	LoRA	32	64	**1.13**	**0.903**	**0.3037**	**0.2307**	**0.848**	**0.9007**	2.04	3.31
	QLoRA	8	16	1.16	0.837	0.3361	0.2639	0.872	0.8483	3.08	0.52
	QLoRA	16	32	1.15	0.883	0.3210	0.2536	0.835	0.8435	3.11	0.52
	QLoRA	32	64	1.15	0.849	0.3177	0.2533	0.871	0.8427	3.15	0.52
ChatGPT2	LoRA	8	16	1.53	0.875	0.2473	0.2261	0.805	0.8269	0.62	0.23
	LoRA	16	32	1.41	0.866	0.2847	0.2547	0.859	0.8362	0.64	0.23
	LoRA	32	64	1.29	0.855	0.2461	0.2168	0.844	0.8260	0.68	0.23
	QLoRA	8	16	1.48	0.833	0.2602	0.2376	0.828	0.8382	0.89	0.04
	QLoRA	16	32	1.32	0.809	0.2221	0.2017	0.793	0.8204	0.89	0.04
	QLoRA	32	64	1.26	0.807	0.2502	0.2066	0.806	0.8322	0.83	0.04
Qwen	LoRA	8	16	1.14	0.866	0.3100	0.2536	0.899	0.8420	1.53	6.84
	LoRA	16	32	1.15	0.860	0.2917	0.2463	0.834	0.8436	1.56	6.84
	LoRA	32	64	1.15	0.885	0.3178	0.2425	0.870	0.8463	1.57	6.84
	QLoRA	8	16	1.15	0.834	0.2911	0.2209	0.839	0.8485	2.52	2.29
	QLoRA	16	32	1.15	0.832	0.2987	0.2320	0.796	0.8463	2.46	2.29
	QLoRA	32	64	1.15	0.852	0.3347	0.2600	0.844	0.8502	2.47	2.29

QLoRA, DeepSeek performance decreased slightly, with precision ranging from 0.837 to 0.883 and F1 scores between 0.835 and 0.872. Its perplexity increased marginally to the 1.15-1.16 range, reflecting the expected loss of expressiveness due to quantization.

ChatGPT2 showed moderate performance across all metrics. Under the tuning of LoRA, accuracy values of 0.875, 0.866, and 0.855 and F1 scores of 0.805, 0.859, and 0.844 were achieved. Perplexity values ranged from 1.29 to 1.53, notably higher than DeepSeek and Qwen, suggesting less fluent generation and a tendency toward more generic or uncertain phrasing. Under QLoRA, the performance of ChatGPT2 further decreased, with accuracy dropping to 0.833, 0.809, and 0.807 and F1 scores to 0.828, 0.793, and 0.806. Perplexity also slightly improved to values between 1.26 and 1.48, indicating a possible trade-off in generation quality toward shorter or simpler completions.

Qwen displayed balanced and reliable performance, combining strong alignment and semantic clarity. Under LoRA, it achieved accuracy values of 0.866, 0.860, and 0.885 and F1 scores of 0.899, 0.834, and 0.870. The ROUGE and BERTScore results were competitive, with a consistently low perplexity (1.14-1.15). With QLoRA, Qwen's performance declined slightly but remained robust. Accuracy values ranged from 0.834 to 0.852, and F1 scores from 0.796 to 0.844. Perplexity was unaffected and stayed close to 1.15, showcasing Qwen's syntactic fluency even under quantization.

In addition to advisory relevance and linguistic quality, each model was also evaluated based on inference time and memory footprint. DeepSeek's LoRA models required 3.31GB and had an average inference time of about 2 seconds. Its QLoRA versions lowered memory requirements to 0.52GB, but this came with a slight increase to 3.1 seconds for inference time. Meanwhile, ChatGPT2 required the least amount of resources, in which the QLoRA models only needed 0.04GB and had an inference time of under 0.9 seconds, making it the most efficient though with the least performance. LoRA variants of Qwen were the most memory-intensive (6.84GB), requiring about 1.5 seconds for inference, while QLoRA dropped these figures to 2.29 GB, with a prediction time of about 2.5 seconds.

Lastly, r and α were hyperparameters having a measurable but subtle effect on performance. Larger values (for example, $r = 32$, $\alpha = 64$) counted for increased F1 Score and BERTScore under LoRA, pointing towards a larger representational capacity. This effect was most visible in DeepSeek, where the highest accuracy (0.903) and lowest perplexity (1.13) occurred under the highest configuration. Under QLoRA, the influence of these hyperparameters was diminished but still detectable, particularly in models like Qwen where the larger configuration preserved semantic and structural quality despite quantization.

In summary, DeepSeek was the top performer in terms of behavioral coverage, generation quality, and fluency, especially under LoRA fine-tuning. Qwen offered a robust middle ground, excelling in both accuracy and efficiency, particularly under QLoRA. ChatGPT2, while less consistent in behavioral alignment, delivered highly efficient inference and small memory requirements. These results imply that not only model architecture but the choice of a fine-tuning strategy and configuration also matter in view of the intended deployment environment.

4.5. Use cases

To illustrate the practical functionality and user relevance of the GUIDE framework, three representative household scenarios are presented. These use cases demonstrate how the best-performing LLM (DeepSeek fine-tuned with LoRA at $r = 32$, $\alpha = 64$) generates personalized energy-saving recommendations based on contextual behavioral data extracted from appliance usage. For each scenario, potential energy savings if the user applies the suggested advice were estimated.

Use Case 1: Overuse of Electric Heater

Scenario description The electric heater was activated 4 times in one day, running for a total of 300 min. The usage exhibited high variability with standard deviation equals to 2.8 and the entropy close to 1.2, indicating inconsistent and excessive operation.

Prompt input

```
Device: Electric Heater
Activations per day: 4
Average activation duration: 75\,min
Total active duration: 300\,min
Days since last use: 0
Usage standard deviation: 2.8
Usage entropy: 1.2
```

LLM output *Your electric heater usage today is notably high. To save energy, consider consolidating heating times or using a programmable thermostat. Setting a consistent schedule during off-peak hours can help reduce energy costs and extend device lifespan.*

Estimated energy saving Assuming the heater consumes 2.0 kW when active, daily energy use $= 2.0 \times 5\,\text{h} = 10\,\text{kWh/day}$. By following the advice to reduce usage by 25 % (e.g., through schedule optimization), the user could save 2.5 kWh/day.

Use Case 2: Prescriptive Maintenance on Washing Machine

Scenario description The washing machine was used 5 times in a single day, with an unusually high total runtime of 450 min. This

exceeds the normal operational range and suggests potential overuse, increasing the risk of mechanical wear or overheating.

Prompt input

```
Device: Washing Machine
Activations per day: 5
Average activation duration: 90\,min
Total active duration: 450\,min
Days since last use: 0
Usage standard deviation: 4.1
Usage entropy: 2.0
```

LLM output *Today's washing machine usage is above typical levels. Excessive daily cycles can lead to mechanical strain or overheating. Consider distributing laundry loads over multiple days or using full-load settings to reduce cycle frequency.*

Estimated energy saving A washing machine cycle typically consumes 0.6 kWh. Reducing from 5 to 3 daily cycles yields a saving of 1.2 kWh/day, resulting in a 24 % energy reduction. More importantly, spreading usage reduces long-term wear and potential repair costs.

Use Case 3: Operational Consistency for Dishwasher

Scenario description The dishwasher has shown irregular usage patterns, with large gaps followed by back-to-back cycles. This inconsistency affects energy optimization and can reduce appliance lifespan due to inefficient start-stop conditions.

Prompt input

```
Device: Dishwasher
Activations per day: 2
Average activation duration: 55\,min
Total active duration: 110\,‘min
Days since last use: 4
Usage standard deviation: 5.2
Usage entropy: 2.3
```

LLM output *Dishwasher use appears inconsistent. Frequent shifts between inactivity and high usage may reduce long-term efficiency. Aim for a regular schedule and run only full loads to optimize water and energy use while maintaining appliance health.*

Estimated energy saving Each cycle uses about 1.2 kWh. By transitioning to a full-load schedule every other day, energy use can drop from 2.4 kWh/day to 1.2 kWh/day, a 50 % energy reduction. Consistent usage also reduces thermal stress on internal components.

These use cases demonstrate the GUIDE framework's ability to provide appliance-specific, behavior-aware recommendations that are not only linguistically personalized but also capable of delivering tangible energy savings.

5. Conclusion

In this paper, a hybrid AI framework (GUIDE) that combines behavioral modeling, device-level disaggregation, and large language model (LLM) guidance to support energy-efficient appliance usage in residential settings was introduced. The system utilizes smart plug data along with user behavior profiles to extract temporal and statistical features that improve classification accuracy and context understanding of appliance usage behavior. The personalized advisory engine is then powered by a fine-tuned LLM (such as DeepSeek fine-tuned with LoRA/QLoRA), generating natural language suggestions tailored specifically for real-life scenarios in the household.

Experimental results on the REFIT dataset revealed that integrating behavioral modeling significantly improves device detection accuracy across multiple machine learning classifiers, with LightGBM achieving an F1-score of 0.913. The advisory component showed strong fluency, semantic relevance, and operational practicality, with DeepSeek outperforming other LLMs in both linguistic quality and behavioral alignment. The use case simulations illustrated GUIDE's applicability to real-world conditions by providing meaningful recommendations and estimating energy savings for behavioral adjustments of sporadically used devices.

Regarding the future work the integration of the system with edge computing platforms and mobile devices to support real-time delivery of energy-saving advice would be challenging. Moreover, while the present work focuses on technical evaluation, future studies will incorporate user-centered evaluations, such as surveys and controlled experiments, to assess the perceived usefulness and behavioral impact of GUIDE-generated advice. In terms of generalizability, GUIDE has been validated on the REFIT dataset, which provides multiple households with varied appliance routines. This confirms applicability across different household usage patterns within that dataset. However, broader transferability across climates, tariff structures, and cultural contexts remains an open challenge. Future research will therefore evaluate GUIDE using additional datasets and cross-regional studies to assess scalability and robustness under diverse environmental and socio-economic conditions. Additionally, would be really valuable to enhance the behavioral modeling component to handle multi-user environments, identifying conflicting usage patterns and adapting recommendations accordingly. Another promising direction involves incorporating reinforcement learning techniques to enable dynamic optimization of recommendations based on historical user responses.

CRediT authorship contribution statement

Alexios Papaioannou: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization; **Asimina Dimara:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Data curation; **Stelios Krinidis:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is partially supported by the HARMONISE project funded by the European Union's Horizon 2020 under Grant Agreement No. 101138595.

Appendix

Table 6
Device detection performance per appliance: with and without behavioral features (four decimal digits).

Model	Device	Precision (without)	Precision (with)	Recall (without)	Recall (with)	F1-score (without)	F1-score (with)
Decision Trees (DT)	Fridge	0.8200	0.8854	0.8000	0.8702	0.8100	0.8777
	Washing Machine	0.7500	0.8608	0.7300	0.8456	0.7400	0.8532
	Dishwasher	0.7800	0.8489	0.7600	0.8331	0.7700	0.8410
	Dryer	0.7900	0.8520	0.7700	0.8365	0.7800	0.8442
	TV	0.7700	0.8872	0.7500	0.8710	0.7600	0.8790
	Kettle	0.7400	0.8385	0.7200	0.8221	0.7300	0.8302
	Microwave	0.7500	0.8452	0.7300	0.8297	0.7400	0.8373
	Toaster	0.7300	0.8271	0.7100	0.8110	0.7200	0.8190
	Electric Heater	0.7500	0.8435	0.7300	0.8280	0.7400	0.8357
Random Forest (RF)	Fridge	0.8400	0.9023	0.8200	0.8884	0.8300	0.8953
	Washing Machine	0.7800	0.8745	0.7600	0.8610	0.7700	0.8677
	Dishwasher	0.8000	0.8647	0.7800	0.8505	0.7900	0.8575
	Dryer	0.8000	0.8689	0.7800	0.8542	0.7900	0.8615
	TV	0.7900	0.9028	0.7700	0.8891	0.7800	0.8960
	Kettle	0.7600	0.8505	0.7400	0.8367	0.7500	0.8436
	Microwave	0.7700	0.8570	0.7500	0.8432	0.7600	0.8501
	Toaster	0.7400	0.8351	0.7300	0.8204	0.7350	0.8277
	Electric Heater	0.7700	0.8561	0.7500	0.8418	0.7600	0.8489
AdaBoost	Fridge	0.8300	0.8942	0.8100	0.8807	0.8200	0.8874
	Washing Machine	0.7600	0.8628	0.7400	0.8494	0.7500	0.8561
	Dishwasher	0.7800	0.8545	0.7600	0.8403	0.7700	0.8473
	Dryer	0.7800	0.8577	0.7600	0.8436	0.7700	0.8506
	TV	0.7700	0.8924	0.7600	0.8790	0.7650	0.8857
	Kettle	0.7400	0.8443	0.7200	0.8305	0.7300	0.8373
	Microwave	0.7600	0.8518	0.7400	0.8376	0.7500	0.8447
	Toaster	0.7300	0.8294	0.7100	0.8151	0.7200	0.8222
	Electric Heater	0.7500	0.8483	0.7300	0.8342	0.7400	0.8412
Support Vector Machine (SVM)	Fridge	0.8200	0.8897	0.8000	0.8759	0.8100	0.8827
	Washing Machine	0.7500	0.8587	0.7300	0.8446	0.7400	0.8516
	Dishwasher	0.7700	0.8412	0.7500	0.8274	0.7600	0.8342
	Dryer	0.7700	0.8445	0.7500	0.8307	0.7600	0.8375
	TV	0.7600	0.8886	0.7400	0.8751	0.7500	0.8818
	Kettle	0.7400	0.8381	0.7200	0.8243	0.7300	0.8311
	Microwave	0.7500	0.8450	0.7300	0.8313	0.7400	0.8381
	Toaster	0.7200	0.8192	0.7100	0.8050	0.7150	0.8121
	Electric Heater	0.7400	0.8427	0.7200	0.8289	0.7300	0.8358
XGBoost	Fridge	0.8600	0.9160	0.8400	0.9027	0.8500	0.9093
	Washing Machine	0.7800	0.8809	0.7600	0.8671	0.7700	0.8740
	Dishwasher	0.8000	0.8692	0.7800	0.8553	0.7900	0.8622
	Dryer	0.8000	0.8730	0.7800	0.8590	0.7900	0.8660
	TV	0.7900	0.9089	0.7700	0.8945	0.7800	0.9017
	Kettle	0.7600	0.8539	0.7400	0.8402	0.7500	0.8470
	Microwave	0.7700	0.8612	0.7500	0.8476	0.7600	0.8543
	Toaster	0.7400	0.8403	0.7300	0.8264	0.7350	0.8334
	Electric Heater	0.7600	0.8583	0.7400	0.8447	0.7500	0.8515
LightGBM	Fridge	0.8700	**0.9417**	0.8600	**0.9275**	0.8650	**0.9346**
	Washing machine	0.8000	**0.9183**	0.7800	**0.9045**	0.7900	**0.9113**
	Dishwasher	0.8200	**0.9020**	0.8000	**0.8884**	0.8100	**0.8951**
	Dryer	0.8200	**0.9063**	0.8000	**0.8927**	0.8100	**0.8995**
	TV	0.8000	**0.9543**	0.7800	**0.9410**	0.7900	**0.9476**
	Kettle	0.7800	**0.8945**	0.7600	**0.8807**	0.7700	**0.8876**
	Microwave	0.7900	**0.9101**	0.7700	**0.8962**	0.7800	**0.9031**
	Toaster	0.7600	**0.8889**	0.7500	**0.8750**	0.7550	**0.8819**
	Electric Heater	0.7800	**0.9025**	0.7600	**0.8891**	0.7700	**0.8958**

References

[1] A. Soares, Á. Gomes, C.H. Antunes, Categorization of residential electricity consumption as a basis for the assessment of the impacts of demand response actions, Renewable Sustain. Energy Rev. 30 (2014) 490–503.

[2] C. Sardianos, I. Varlamis, C. Chronis, G. Dimitrakopoulos, A. Alsalemi, Y. Himeur, F. Bensaali, A. Amira, The emergence of explainability of intelligent systems: delivering explainable and personalized recommendations for energy efficiency, Int. J. Intell. Syst. 36 (2) (2021) 656–680.

[3] A. Papaioannou, A. Dimara, C. Papaioannou, I. Papaioannou, S. Krinidis, C.-N. Anagnostopoulos, C. Korkas, E. Kosmatopoulos, D. Ioannidis, D. Tzovaras, Simulation of malfunctions in home appliances' power consumption, Energies 17 (17) (2024).

[4] A. Papaioannou, A. Dimara, I. Papaioannou, A. Stefanopoulou, L. Georgios, I. Michailidis, C. Korkas, C.-N. Anagnostopoulos, E. Kosmatopoulos, S. Krinidis, et al., An innovative methodology for revealing home appliances' consumption patterns to transform energy management and maintenance strategies, in: IFIP International Conference on Artificial Intelligence Applications and Innovations, Springer, 2024, pp. 377–390.

[5] M. Liu, L. Zhang, J. Chen, W.-A. Chen, Z. Yang, L.J. Lo, J. Wen, Z. O'Neill, Large language models for building energy applications: opportunities and challenges, in: Building Simulation, Springer, 2025, pp. 1–10.

[6] L. Zhang, Z. Chen, Opportunities and challenges of applying large language models in building energy efficiency and decarbonization studies: an exploratory overview, (2023) arXiv preprint arXiv:2312.11701.

[7] M. Umair, G.A. Shah, Energy management of smart homes, in: 2020 IEEE International Conference on Smart Computing (SMARTCOMP), 2020, pp. 247–249. https://doi.org/10.1109/SMARTCOMP50058.2020.00054

[8] B. Mataloto, J.C. Ferreira, R.P. Resende, Long term energy savings through user behavior modeling in smart homes, IEEE Access 11 (2023) 44544–44558. https://doi.org/10.1109/ACCESS.2023.3272888

[9] M. Barsanti, S. Yilmaz, C.R. Binder, Informing targeted demand-side management: leveraging appliance usage patterns to model residential energy demand heterogeneity, Energy Build. 321 (2024) 114639. https://doi.org/https://doi.org/10.1016/j.enbuild.2024.114639

[10] S.M. Hoosh, I. Kamyshev, H. Ouerdane, Enhancing non-intrusive load monitoring with features extracted by independent component analysis, 2025. https://arxiv.org/abs/2501.16817.

[11] F. Yu, Z. Wang, X. Zhang, M. Xia, DRA-net: a new deep learning framework for non-intrusive load disaggregation, Front. Energy Res. 11 (2023) 1140685.

[12] S. Liu, Z. Xie, Z. Hu, Optimizing smart home appliance energy monitoring using factorial hidden Markov models for internet of behavior, J. Build. Eng. 97 (2024) 110732. https://doi.org/https://doi.org/10.1016/j.jobe.2024.110732

[13] M. Ma, S. Zhang, J. Liu, R. Yan, W. Cai, N. Zhou, J. Yan, Building floorspace and stock measurement: a review of global efforts, knowledge gaps, and research priorities, Nexus (2025).

[14] T. Wang, M. Ma, N. Zhou, Z. Ma, Toward net zero: assessing the decarbonization impact of global commercial building electrification, Appl. Energy 383 (2025) 125287.

[15] F. Santos Fialho, S. Scuri, L. Pereira, N.J. Nunes, Virtual assistants for energy efficiency: real world tryouts, in: Proceedings of the 2023 ACM Conference on Information Technology for Social Good, 2023, pp. 299–306.

[16] A. Suresan, S.S. Mohan, M.P. Arya, V. Anjana Gangadharan, P.V. Bindu, A conversational AI chatbot in energy informatics, in: Proceedings of International Conference on Intelligent Computing, Information and Control Systems: ICICCS 2020, Springer, 2021, pp. 543–554.

[17] T. He, F. Jazizadeh, L. Arpan, Ai-powered virtual assistants nudging occupants for energy saving: proactive smart speakers for HVAC control, 2022 50, (4), 394–409.

[18] G. Gamage, N. Mills, P. Rathnayaka, A. Jennings, D. Alahakoon, Cooee: an artificial intelligence chatbot for complex energy environments, in: 2022 15th International Conference on Human System Interaction (HSI), IEEE, 2022, pp. 1–5.

[19] G. Gamage, S. Kahawala, N. Mills, D. De Silva, M. Manic, D. Alahakoon, A. Jennings, Augmenting industrial chatbots in energy systems using Chatgpt generative AI, in: 2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE), IEEE, 2023, pp. 1–6.

[20] F. Michelon, Y. Zhou, T. Morstyn, Large language model interface for home energy management systems, 2025. https://arxiv.org/abs/2501.07919.

[21] M. Barsanti, S. Yilmaz, C.R. Binder, Informing targeted demand-side management: leveraging appliance usage patterns to model residential energy demand heterogeneity, Energy Build. 321 (2024) 114639.

[22] Z. Pu, Y. Huang, M. Weng, Y. Meng, Y. Zhao, G. He, Enhancing non-intrusive load monitoring with weather and calendar feature integration in DAE, Front. Energy Res. 12 (2024) 1361916.

[23] A. Papaioannou, A. Dimara, S. Krinidis, C.-N. Anagnostopoulos, D. Ioannidis, D. Tzovaras, Advanced proactive anomaly detection in multi-pattern home appliances for energy optimization, Internet Things 26 (2024) 101175.

[24] S. Dash, N.C. Sahoo, Electric energy disaggregation via non-intrusive load monitoring: a state-of-the-art systematic review, Electr. Power Syst. Res. 213 (2022) 108673.

[25] J. Chavat, J. Graneri, S. Nesmachnow, Household energy disaggregation based on pattern consumption similarities, in: Ibero-American Congress of Smart Cities, Springer, 2019, pp. 54–69.

[26] U. S. Department of Energy, Energy saver guide: tips on saving money and energy at home, Technical Report, U.S. Department of Energy, 2022. https://www.energy.gov/sites/default/files/2022-08/energy-saver-guide-2022.pdf.

[27] Carbon Trust, Better business guide to energy saving, Technical Report, Carbon Trust, 2012. Accessed: 2025-09-04, https://ctprodstorageaccountp.blob.core.windows.net/prod-drupal-files/documents/resource/public/Better-Business-Guide.pdf.

[28] Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products, Regulation, European Parliament and Council of the European Union, 2024. Official Journal of the European Union, L 181; Accessed: 2025-09-04, https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L_2024_01781.

[29] C. Beckel, L. Sadamori, S. Santini, Automatic socio-economic classification of households using electricity consumption data, in: Proceedings of the fourth international conference on Future energy systems, 2013, pp. 75–86.

[30] E.J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, et al., Lora: low-rank adaptation of large language models, ICLR 1 (2) (2022) 3.

[31] Y. Xu, L. Xie, X. Gu, X. Chen, H. Chang, H. Zhang, Z. Chen, X. Zhang, Q. Tian, QA-LORA: quantization-aware low-rank adaptation of large language models, (2023) arXiv preprint arXiv:2309.14717.

[32] D. Murray, L. Stankovic, V. Stankovic, An electrical load measurements dataset of United Kingdom households from a two-year longitudinal study, Sci. Data 4 (1) (2017) 1–12.